Quantum Mechanics Assignment-I
CY1001
Jan-May 2026

1. Which of these functions is acceptable as a wave function?
   (a) $\frac{1}{x}(0 \leq x \leq \infty)$
   (b) $\sin(x)\ (-\infty \leq x \leq \infty)$
   (c) $e^{-x}\ (0 \leq x \leq \infty)$
   (d) $e^{-x^2}(-\infty \leq x \leq \infty)$

2. Which of the operators are nonlinear-
   (a) $\frac{\partial^2}{\partial x^2}$
   (b) $(\quad)^2$
   (c) $\sqrt{()}$
   (d) $3x^2$

3. Consider two operators $\hat{A}$ and $\hat{B}$. Which of the following statements are true regarding their commutativity?

   a. If $[\hat{A}, \hat{B}] = 0$, then $\hat{A}$ and $\hat{B}$ must share the same eigenstates.
   b. If $[\hat{A}, \hat{B}] = 0$, then $\hat{A}$ and $\hat{B}$ will have identical eigen values.
   c. If $[\hat{A}, \hat{B}] = 0$, it implies that the measurement is independent of the order in which the operators act on the function.
   d. If $[\hat{A}, \hat{B}] = 0$, the measurements of $\hat{A}$ and $\hat{B}$ will not disturb each other

4. Determine whether or not the following pairs of operators commute.

$\hat{A}$	$\hat{B}$
$x^2\frac{d}{dx}$	$\frac{d^2}{dx^2}$
$\frac{d}{dx} - x$	$\frac{d}{dx} + x$
$\frac{d^2}{dx^2} - x$	$\frac{d^2}{dx^2} + x$

5. Which of these functions are eigenfunctions of the operators $\frac{d}{dx}$ as well as $\frac{d^2}{dx^2}$
   (a) e^{-ax^2} where a is constant
   (b) $\cos(bx)$ where b is constant
   (c) e^{ikx} where k is constant
   (d) $\sin(bx)$ where b is constant

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6. Show that the function f(x) = (8 e^{5x} + 9 e^{-5x}) an eigen function of the operator d^2/dx^2 . What is the eigen value?

7. For a quantum mechanical system, $\Psi = cos\left(\frac{\pi x}{L}\right)$ for $-\frac{L}{2} < x < \frac{L}{2}$. Normalize Ψ.

8. Consider a particle in a quantum state Ø that is the superposition of two eigen functions of energy $\emptyset_1$ and $\emptyset_2$ with energy eigen values E_1 and E_2:

   $\emptyset = C_1\emptyset_1 + C_2\emptyset_2$

What is the probability of measuring E_1 or E_2? What is the average energy?

9. The speed of an electron is 995 km/s. If the uncertainty in its momentum is to be reduced to 0.01 percent, what uncertainty in its position must be tolerated?

10. A particle is confined to a 3D box with sides of length $L_x = L$, $L_y = 2L$, and $L_z = 4L$. Calculate energy of the first five successive energy levels. Which state(s) is/are degenerate?

11. Calculate the probability of finding a particle of mass *m* in a 1D box of length L between the length L/4 and L/2 in state n = 1.

12. For a particle in 1-D box of length L, show that the Heisenberg's uncertainty principle with respect to position and momentum holds true.